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Paleomagnetism and rock magnetism of Early Silurian Dunn Point volcanics, Avalon Zone, Nova Scotia

Maurice K.-Seguin and K.V. Rao *

Paleomagnetic Laboratory, Faculty of Science and Engineering, Université Laval, Québec, G1K 7P4 (Canada)

E.R. Deutsch

Geomagnetic Research Laboratory, Department of Earth Sciences, Memorial University of Newfoundland, St. John's, Newfoundland A1B 3X7 (Canada)

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Paleomagnetic results have been obtained from the subaerial volcanics (andesites, rhyolites, ignimbrites) from the Dunn Point Formation exposed in the Antigonish Highlands of the Avalon Zone, Nova Scotia. An Early Silurian age of 430 ± 15 Ma is assigned to these volcanic rocks. The age is in agreement with the fossil evidence from conformably overlying sedimentary rocks. The volcanic units were tightly folded and metamorphosed to sub-greenschist facies during one major and one secondary phase of deformation probably associated with the Mid–Late Devonian Acadian orogeny.

This paleomagnetic study is based on 45 samples collected from nine sites. The site-mean directions of the natural remanent magnetization (NRM) make a large deviation from the present Earth's field with NRM intensities ranging from 10^{-2} to 10 Am^{-1}. The most commonly observed remanence direction, A ($D = 153°$, $I = -28°$), obtained from both alternating fields (30–75 mT) and thermal (500–560 °C) treatments yields a corresponding direction A' ($D = 0°$, $I = -63°$) after correction for geological tilt. A large scatter in direction (up to $= 60°$) is observed in A' and the fold test is not significantly positive. A second and less common remanence component, B ($D = 146°$, $I = +54°$), which presumably postdates folding, is also observed; this component is mainly isolated by thermal treatment in the 540–630 °C range. Room-temperature isothermal experiments and optical examination of polished sections indicate the presence of magnetite and hematite as the carriers of both A and B directions. The corresponding reversed paleomagnetic poles fall at A (163° E, 52° N) and B (146° E, 4° N) in close agreement with cratonic North American Upper Jurassic and Late Devonian poles, respectively. In the Mid–Late Devonian time interval, the Avalon Zone was situated slightly to the south but close to the North American eastern margin.

1. Introduction

The Silurian paleomagnetic data base for the circum-Atlantic continents is scarce. Brock (1981) made a summary of the paleomagnetic data from Africa for the Lower Paleozoic. No Silurian paleomagnetic data are listed for northern and western Africa. According to Brock, problems of exposure, deformation and metamorphism have made the Lower Paleozoic of Africa unattractive for paleomagnetic studies. In their review of pre-Carboniferous paleomagnetism of Europe north of the Alpine orogenic belt, Briden and Duff (1981) noted that in spite of the fact that the Paleozoic rocks have been remagnetized in the Hercynian

* Now at: Haileybury School of Mines, Haileybury, Ontario, P0J 1K0 Canada.

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orogeny, the primary signatures were probably extracted from half-a-dozen Caledonian rock units mainly from the United Kingdom and Norway. The estimated paleopole for this time interval is about 270° E and 10° S. The number of Silurian paleopoles from the North American craton (e.g., van der Voo, 1981) is smaller than for northwestern Europe. Their reliability is doubtful in view of frequent Carboniferous–Permian overprint on the primary remanence. The presumably reliable paleopole position for the Silurian of North America is located at approximately 120° E, 30° N (Lynnes and van der Voo, 1984). More recently, Seguin and Petryk (1984, 1985) carried out a paleomagnetic study of the Lower Silurian platform sequence of Anticosti Island and obtained paleopoles located further east (140–160° E) and closer to the equator (10–15° N). To our knowledge, the Silurian pole position of stable North America is still insufficiently defined. The paleolatitudinal data available from the different paleomagnetic studies indicate that North America (van der Voo, 1981) and Baltica (Briden and Duff, 1981) were situated close to the equator while northwestern Africa (Brock, 1981) and possibly Hercynian Europe (Perroud et al., 1984) were closer to the pole. Baltica and North America collided along an E–W direction to produce the so-called Taconic phase of the Caledonian–Appalachian orogenic belt by Late Ordovician to Mid-Silurian time.

The Dunn Point Formation was folded and metamorphosed at the same time as the Devonian sedimentary rocks of the Arisaig Group. Consequently, the Dunn Point Formation was not involved in the Caledonian orogeny but probably in both the Acadian and Alleghenian (Hercynian) orogenies and most certainly in the first one (Acadian). The cause(s) of these orogenies is (are) difficult to explain. Many hypotheses were proposed to explain the Acadian orogeny (e.g., collision of Gondwana with North America along a SW–NE direction; collision of a portion of northern Europe, namely the northern British Isles, with North America; succession of suspect terranes accreted to the eastern margin of the North American craton at specific times, etc...). Thus, the mechanism responsible for the incorporation of the Avalonian Terrane to the eastern border of the North American continent and the provenance of this terrane are still uncertain. Different faunas between Avalon terrane(s) and cratonic North America and similar lithologies between Avalon and northern Europe and/or North Africa suggest that Avalon is a suspect and probably displaced terrane (Schenk, 1971, 1975; Rast and Skehan, 1983; Skehan and Rast, 1983; Williams and Hatcher, 1983; O'Brien et al., 1983).

Preliminary paleomagnetic results of the Dunn Point Formation were published by Seguin et al. (1981b). A detailed paleomagnetic study of the Dunn Point Formation was published by van der Voo and Johnson (1985). Their results indicate high Late Ordovician–Early Silurian paleolatitudes and they concluded to an important post-Ordovician displacement of Avalon with respect to cratonic North America. These authors correlated the high paleolatitudes in the Dunn Point Formation with those obtained in northern Africa and Hercynian Europe.

The present paleomagnetic investigation has four objectives:

(1) it constitutes a detailed paleomagnetic study of the Early Silurian Dunn Point volcanics and aims at obtaining well-dated paleomagnetic data;

(2) it is part of a major program aimed at studying paleomagnetism of the Avalon Zone in New Brunswick, Nova Scotia and Newfoundland (Rao et al., 1981) to determine whether this mini-continent is composed of a mosaic of smaller blocks or one single terrane;

(3) it is used for a verification of the hypothesis of a Silurian sea by comparison of the North American cratonic and Avalon poles of approximately the same age. This hypothesis was first formulated by Wilson (1966) and McKerrow and Ziegler (1972); and

(4) it is an attempt to estimate the amount of displacement of this terrane with respect to the existing larger plates in this time interval (e.g., Africa, North America and northwestern Europe).

2. Geological situation and sampling sites

Figure 1 shows a location map of the area investigated. It is situated in the northeastern part

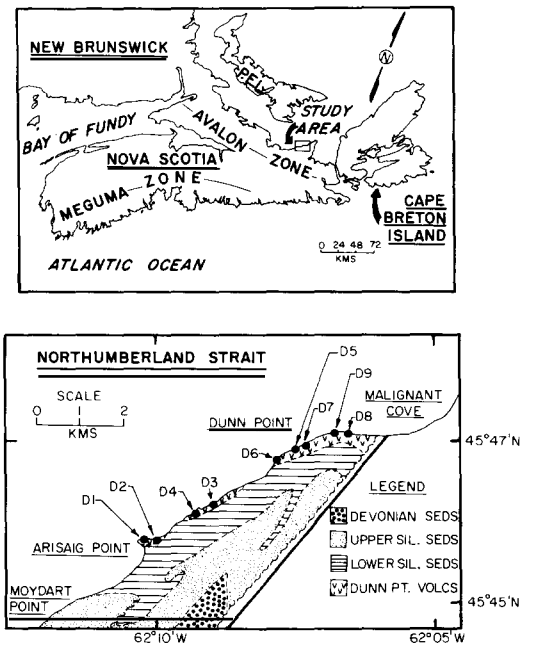


Fig. 1. The top diagram shows the approximate boundary of zones and fragments of ancient plates of the northern Appalachians. These zones are: (1) cratonic margin situated to the north; (2) Andean margin of Iapetus (central and northern New Brunswick and northwards); (3) Avalon terrane; and (4) Meguma terrane. The bottom diagram shows a location and geological map (modified after Benson, 1974) of the area investigated which is situated in the northeastern part of Nova Scotia, Canada, south of the Northumberland Strait. Nine of the 11 sampling sites (those which could be demagnetized) are indicated on this diagram.

of Nova Scotia, Canada, south of the Northumberland Strait (62°20′W, 45°46′N). The volcanosedimentary rock units of this region were described by Benson (1974), Boucot et al. (1974) and Keppie et al. (1978). The Dunn Point volcanics are a bimodal suite, some 200 m thick consisting of basalt, rhyolite flows, volcanic tuffs and breccias, and ignimbrites. The individual flows are 1–5 m thick. Red soils interpreted as laterites (equatorial paleogeography) (Boucot et al., 1974) are occasionally interstratified with the volcanic units. This suggests a subaerial deposition of the volcanic units. The fact that the tops of the flows are more oxidized and that laterite xenoliths are incorporated in some of the flows may indicate that the hematite found in the beds or flows is primary. However, Keppie et al. (1979) state that the volcanics contain almost totally secondary minerals (including the oxides). The Dunn Point volcanics are overlain by the McGillivray Brook Formation and the fossiliferous Lower Silurian Arisaig Group above it. A Rb/Sr whole rock isochron obtained by Fullager and Bottino (1968) on the Dunn Point volcanics gave an age of 430 ± 15 Ma, i.e., Early Silurian. Northeast of Arisaig Brook (≃ 1.5 km SW of Arisaig Point (Fig. 1)), the northern southward facing limb of the Arisaig synclinal fold (in Devonian sediments) is generally overturned and the southern, northward facing limb of the fold is gently inclined to the west or northwest (Boucot et al., 1974). The regional fold is gently plunging to the southwest. The structural geology of the Arisaig area is very complex; according to Boucot et al. (1974), the rocks of this area bear evidence of five phases of deformation and numerous vexing structural problems. Deformation took place at the same time or after the deposition of the upper formation (Knoydart; Lower Devonian) of the Arisaig Group and it is correlated with the Acadian orogeny (Benson, 1974; Boucot et al., 1974). The Dunn Point volcanics have been metamorphosed under greenschist facies and the primary mineral assemblages have been almost completely destroyed (Keppie et al., 1979). Silurian volcanic rocks in the northern Appalachians generally occur in two belts: a northern belt extending from Chaleurs Bay through Quebec, New Brunswick and northern Maine and a southern belt in coastal Maine and southern New Brunswick. The Silurian volcanic rocks in northern Nova Scotia may be an extension of the southern belt. Both belts of volcanic rocks have generally been attributed to calc-alkaline island arc volcanism associated with the closing of the Iapetus Ocean. Keppie et al. (1978) have a divergent opinion; they think that the Dunn Point is of within plate origin, being extruded in a non-orogenic extensional environment related to continental rifting. Eleven sites were sampled but only nine (45 samples, 132 specimens) could be used in the present study; their location is indicated in Fig. 1. Preliminary paleomagnetic results of the Dunn Point Formation were published by Seguin et al. (1981b).

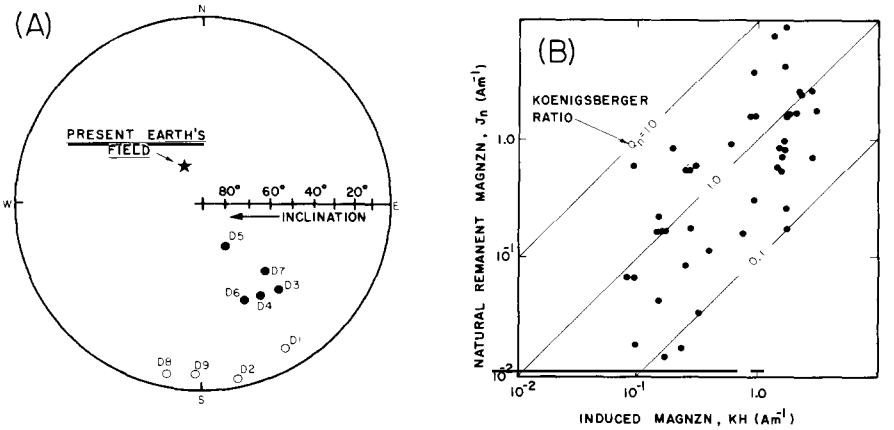


Fig. 2. (A) shows the NRM-site mean directions for the Dunn Point volcanics (in-situ). The NRM directions are generally far away from the present Earth's field (PEF). (B) shows that the NRM intensities (J_n) and the Koenigsberger ratios (Q_n) range from 2×10^{-2} to 8 Am^{-1} and 0.1 to 9, respectively (mean 0.8). The values of J_r and Q_n are very variable from site to site and sometimes within a site.

3. NRM directions and intensities

In-situ NRM directions are most commonly oriented in the SE and south quadrants (Fig. 2); only five specimens coincided with the present Earth's field (PEF). The inclinations are either intermediate positive (five sites) or shallow negative (four sites). NRM intensities (J_r) and Koenigsberger's ratio (Q_n) range from 2×10^{-2} to 8 Am^{-1} and 0.1 to 9 (mean close to unity).

4. Magnetic properties

Data from four types of observations were obtained on 10 samples of volcanics. These are: (1) acquisition of isothermal remanence (IRM) at room temperature; (2) low-field hysteresis (LFH, Rayleigh loops) in 1 mT peak field; (3) high-field hysteresis (HFH) at 20 °C and −196 °C in 120 mT peak field; and (4) susceptibility vs. temperature curves (k–T) obtained in 0.03 mT peak field in the temperature range: -200 to 700 °C. Measurements are in air. For apparatus and measurement details, refer to Seguin et al. (1981a) for experiment 1 and Radhakrishnamurty and Deutsch (1974), Deutsch and Pätzold (1976) for granulometric experiments 2, 3 and 4. The IRM acquisition curves (Fig. 3) did not reach saturation until fields as high as 80 mT were reached, which is typical of hematite, maghemite or (pseudo) single-domain structures of magnetite; also, the corresponding backfield curves suggest the presence of hematite (maghemite) with high values of 20–30 mT and also magnetite (based on sharp inflections) with low values of HCR (5–10 mT). The granulometric experiments (Fig. 4) show no Rayleigh loop. At 20 °C, the high-field coercivity is about 17 mT, increasing to 24–36 mT at −196 °C. The J_r/J_{max} ratio at 20 °C is low (0.20), increasing only slightly at −196 °C. The k–T curves are moderately reversible, suggesting large magnetic grain, and show the three features: isotropic point peak, sharp drop at close to Curie point and relatively temperature independent trend between these features, strongly diagnostic of multi-domain magnetite. The absence of a Rayleigh loop is compatible with this interpretation, but the high-field hysteresis results are not. Multi-domain magnetite should give narrow loops, expanding only by a factor of two or so at −196 °C (Radhakrishnamurty and Deutsch, 1974). The high-field hysteresis results are more indicative of cation-deficient magnetite (i.e., oxidized single-domain grains) of high stability (Radhakrishnamurty and Deutsch, 1974). The samples investigated have

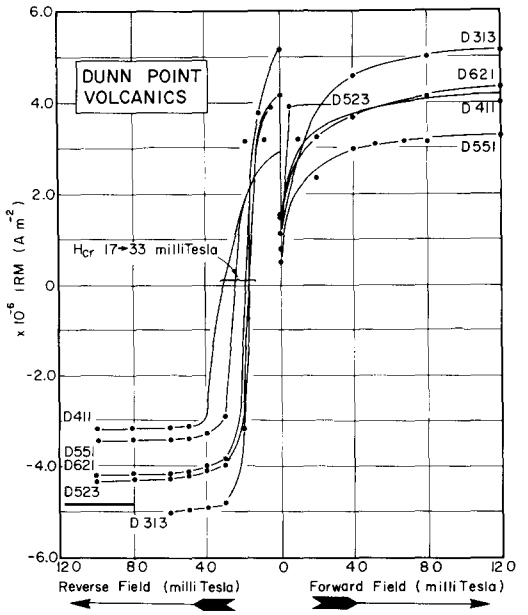


Fig. 3. Acquisition of isothermal remanence (Am^{-2}) at room temperature in forward and reverse field increments up to 0.14 T for five typical specimens (as indicated) from the Dunn Point volcanics. The intersections of curves in the reverse field axis correspond to the values of coercivity of remanence (H_{cr}) as indicated.

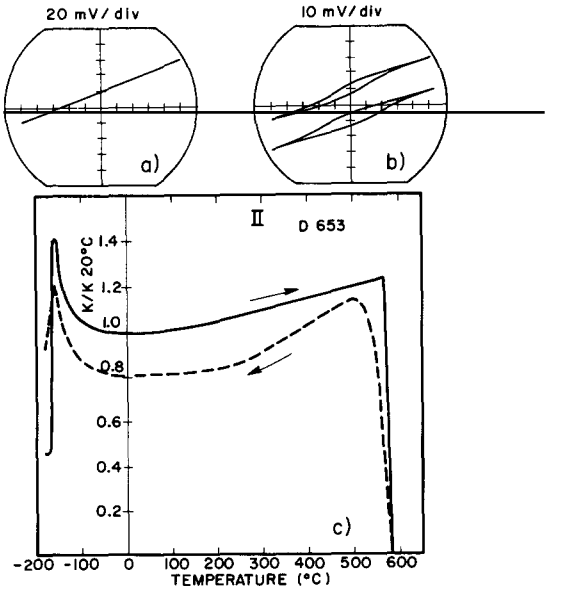


Fig. 4. (a) Low-field hysteresis loops obtained in 1 mT peak on Rayleigh loop tracer (drawings of oscilloscope photo displayed on upper left side), (b) high-field hysteresis loops. These are similar to saturated hysteresis loops, but H_{max} value (120 mT) falls short of saturation, so that coercivity (H_{cr}) values are somewhat smaller than for the saturated case (drawings of oscilloscope photograph are displayed on the upper right side), (c) susceptibility vs. temperature ($k-T$) curves in 0.03 mT peak field on alternating current balance. Normalized heating and cooling curves are traced on the lower part of the diagrams. Two typical samples were chosen (D 653 to the left and D 341 to the right).

a soft (multi-domain) and a hard (cation deficient) component, the former showing up in the $k-T$ curves and the latter in the high-field hysteresis loops. The multi-domain component is probably quite unoxidized, since cation-deficient magnetite does not show a low-temperature transition (Deutsch and Pätzold, 1980). There is no reason that a two component granulometry should not exist; in fact the above observations interpreted as a multi-domain-cation-deficient mixture are very similar to those found in Murthy et al. (1976) and were similarly interpreted there.

The optical examination of many polished sections indicated the presence of predominantly large grains of pure magnetite often in cubic form and a few grains of corroded titanomagnetite. Occasional grains of hemoilmenite are also present. Most commonly, the magnetite is rimmed with hematite. Hematite is also found as thin laths in fractured crystals of magnetite. These observations and others of the same kind show that most of the hematite is secondary (alteration product of magnetite). The amount of magnetite is much more important than that of hematite in all the polished sections.

5. Analysis of paleomagnetic directions

All Silurian volcanic specimens were AF demagnetized by steps of 5 mT from 5 to 100 mT or thermally cleaned by steps of 100 °C at low temperature and 20 °C at high temperature from 100 to 685 °C. At least two and most probably three components of magnetization were isolated. Component A has a SSE, shallow negative direction and is found in all nine sites which could be

demagnetized with success (Fig. 5A, Table I). Component *B* is present in six of the nine sites; it is oriented SSE but with a positive intermediate inclination (Fig. 5b, Table I). Even though less common, component *C* is also present in six of the nine sites. The direction of this last component is WSW with a steep negative inclination (Table I).

Component *A* is characterized by an unblocking temperature (T_{ub}) of 590–640 °C (Fig. 6) and high median destructive fields (Fig. 7a) ranging from 25 to 100 mT (generally 60–70 mT). Component *A* is predominant on sites 1, 2, 8 and 9 (Table I); its memory carrier is magnetite and/or hematite and magnetite. Component *B* has a lower T_{ub} (540–570 °C; generally 555 °C) than component *B* and a much lower median destructive field (5–40 mT; generally 15 mT). Magnetite is the magnetic carrier of this component. Component *B* is mainly found on sites 3, 5, 6 and 7. Hematite is not always the only memory carrier of component *A* (Fig. 7b); there are exceptions to this rule; for instance, in sample 5 of site 7 and sample 3 of site 9, component *B* is present in hematite instead of magnetite. Similarly, magnetite is occasionally the only memory carrier of component *A* (e.g., sample 3 of site 4 and sample 4 of site 6). Still, both components (*A* and *B*) are rarely found in the same samples, let alone the same specimen; exceptions to this rule are samples 1, 2 and 3, 4, 3 of sites 3, 4, 6 and 9, respectively. Component *C* was detected in only one sample on each of the six sites where it was isolated. It is thus difficult to decide whether this is a true component or not. Except for sample 1 of site 8, the magnetic memory of this component is maghemite and hematite (T_{ub} 625 °C, $H_c \geqslant 65$ mT).

In all cases, it is noteworthy that the direction of magnetization does not appreciably change from NRM to high temperature or high fields of demagnetization. Approximately 15% of the specimens were definitely unstable in the course of demagnetization. These coincided quite commonly with the ones which retained a lot of gas in the vesicles of the rock samples and eventually exploded on heating.

6. Interpretation of the results

The Dunn Point volcanics are generally steeply dipping; however, small variations in the strikes and dips can be used for a paleomagnetic fold test. This fold (tilt) test was performed on all three

TABLE I

List of thermally and AF demagnetized site-mean directions

Site	N'	n'	Strike (°)	Dip (°)	Polarity	Component *A*							
						$N(n)$	In-situ				Tilt-corrected		
							D (°)	I (°)	α_{95} (°)	k	D (°)	I (°)	k
1	5	17	57	55SE	U	4(14)	145	−45	27.8	12	335	−80	
2	5	14	71	68SE	U	4(12)	157	−45	14.8	38	348	−67	
3	6	13	62	95NW	O	2(5)	140	−24	–	–	357	−59	
4	7	21	72	71SE	U	4(11)	155	−23	26.1	14	39	−82	
5	6	17	71	82NW	O	1(3)	149	−10	–	–	15	−68	
6	5	18	75	77NW	O	2(7)	149	−15	–	–	15	−58	
7	7	16	74	76NW	O	3(8)	134	−18	29.1	19	27	−47	
8	6	19	78	84NW	O	4(11)	177	−41	30.5	10	339	−41	
9	6	20	81	81NW	O	5(17)	175	−27	6.3	147	345	−54	
10	5	15				Unstable during AF cleaning and exploded during heating							
11	6	18											
			Total and mean										
9	64	188				9(88)	153	−28	11.8	20	000	−63	23

N, N' = number of samples; n, n' = number of specimens; U = upright position; O = overturned; D = declination, I = inclination,

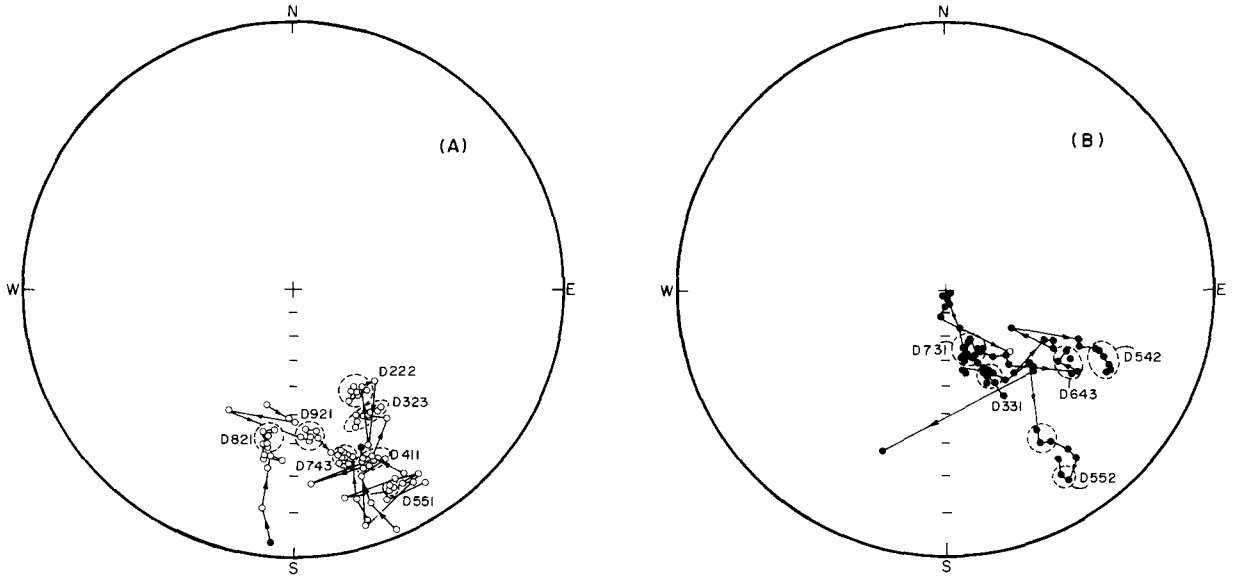


Fig. 5. (A) Response to thermal demagnetization (20–590 °C) of five specimens from four different sites for component *B* (in-situ) of the Dunn Point volcanics and presented in the form of a stereonet. The directional plots are with respect to the present horizontal. (B) response to thermal (20–685 °C) and AF demagnetization (0–100 mT) of seven specimens from seven different sites for component *A* (in-situ) and presented in the form of a stereonet.

components (Table I). In the case of components *B* and *C*, the tilt test is clearly negative; still, it is not quite significantly negative at the 95% confidence level (McElhinny, 1964). Component *A* fails to pass a positive fold test (Table I). Consequently, all three components are secondary. From

Component *B*								Component *C*							
N(*n*)	In-situ				Tilt-corrected			*N*(n)	In-situ				Tilt-corrected		
	D (°)	*I* (°)	α_{95} (°)	*k*	*D* (°)	*I* (°)	*k*		*D* (°)	*I* (°)	α_{95} (°)	*k*	*D* (°)	*I* (°)	*k* (°)
								1(4)	236	−76			310	−34	
3(9)	152	71	18	48	153	−24									
3(10)	158	51	22.6	31	159	−20		1(2)	274	−69			322	−11	
5(13)	145	48	21.4	14	145	−47		1(2)	232	−70			323	−01	
3(11)	141	56	31	11	147	−43									
3(7)	141	49	13.7	82	140	−50		1(3)	259	−48			301	14	
1(3)	141	45	-	-	140	−45		1(4)	284	−63			323	17	
								1(2)	235	−77			339	03	
6(53)	146	54	8.7	81	148	−38	33	6(17)	258	−69	11.2	37	319	−2	14

α_{95} = cone of 95% confidence about the mean (Fisher, 1953), *k* = precision parameter of the mean direction.

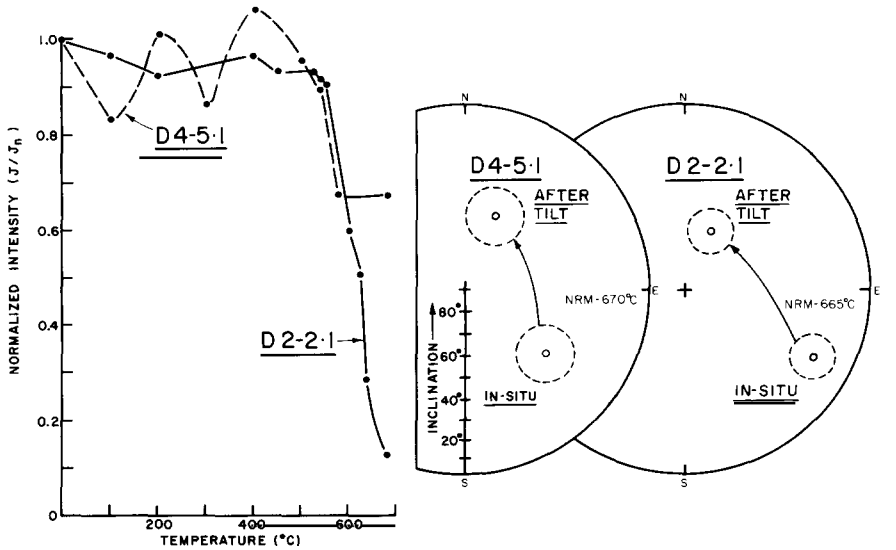


Fig. 6. Examples of 2 thermally cleaned specimens exhibiting the most commonly observed remanence direction (component *A*) in both in-situ and tilt-corrected situations. The largest fraction of the stable memory carrier is magnetite but hematite (T_{ub} 640 ° C) was encountered in some samples. It is noteworthy that the direction of magnetization does not appreciably change from NRM to high temperatures.

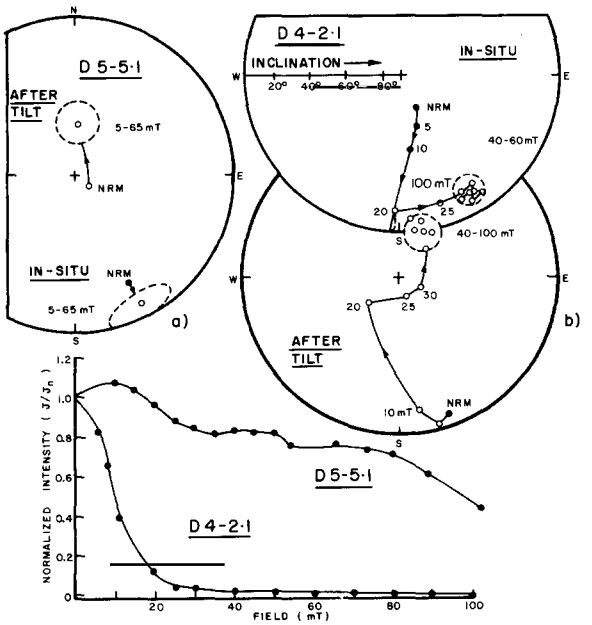


Fig. 7. Typical examples of AF treatment (specimens D4-2.1 and D5-5.1). The normalized intensity curve shows the removal of a superimposed low coercivity component in the 5–20 mT range for specimen D5-5.1 in particular. The stereograms of orientations (both before and after tilt) show that the mean specimens direction of AF and thermally cleaned specimens and samples (e.g., D4-2.1 and D4-5.1 in Fig. 6) is surprisingly comparable when using the minimum dispersion criteria. In these specimens, the magnetization is strong and mainly found in maghemite; it represents the age of alteration of the ignimbrites.

this point, one can speculate on the relative age of these three components.

IRM experiments indicate the general predominance of magnetite over hematite. Specimen 313 which was AF demagnetized contained both *A* and *B* components. Component *B* persisted up to 50 mT and at this field strength, the relative intensity was about 1%. Above 50 mT, the *A* component is well isolated and the intensity increased up to 3% at 100 mT. A combination of both IRM and magnetic granulometry experiments show that higher coercivity IRM (25 up to 50 mT) and high-field hysteresis results are indicative of cation-deficient magnetite including some maghemite (e.g., samples D3-1, D3-4, D5-2) while lower coercivity IRM (15–25 mT) and the absence of a Rayleigh loop are characteristic of multi-domain magnetite. As mixed behaviors are quite common, the tagging of a component to a specific

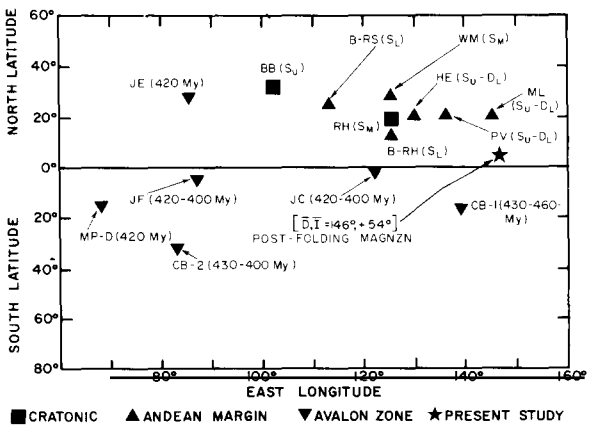


Fig. 8. This diagram shows the Silurian paleopoles obtained in the northern Appalachians. The new result presented in this study (component B in-situ) is shown as a star but it is interpreted as a Middle to Late Devonian event. The two poles from cratonic North America are the Bloomsburg Formation (BB, Roy et al., 1967) and the Rose Hill Formation (RH, French and van der Voo, 1979). The poles from the Andean margin of Iapetus are Botwood sandstone (B-RS, Lapointe, 1979), Botwood rhyolite (B-RH, Lapointe, 1979), Weedon-Gould metavolcanics (PV, Seguin, 1980a), MacKay Lake volcanics (ML, Seguin, 1980a), Hershey Formation (HE, Kent and Opdyke, 1980). The poles from the Avalon terrane(s) are Mascarene Group volcanics (JE, JF, Roy and Anderson, 1981), Mascarene Group intrusives (JC, Roy et al., 1979), Mount Payton diorite (MP-D, Lapointe, 1979), Cape Breton granites (CB-1, CB-2, Rao et al., 1981).

age of magnetization is quite difficult; however, the presence of soft and hard components can be explained.

As both tests on magnetic properties and microscopic observations indicate that hematite and maghemite are alteration products of magnetite and as magnetite is predominantly present in component *B* which was shown to be secondary, one is left with a reliable criterion pointing to a secondary origin for component *A*.

7. Discussion

Based on the study of the magnetic properties and microscopic observations, hematite which represents a small fraction of the opaques is clearly secondary. Thus, oxidation and laterization took place during or after the Acadian orogeny.

Consequently, complicated mineralogy, alteration and magnetic overprinting are the main characteristics of the present paleomagnetic study. As stated earlier, two paleomagnetic studies of the Dunn Point Formation are now available: one (A) by van der Voo and Johnson (1985) and the present study (B). The results and their outcoming interpretation are different in many respects. The paleomagnetic results of study A are non-complex and univectorial; this is not the case in study B. The remanence in study A is carried by hematite while in study B, both magnetite and hematite are memory carriers with magnetite being predominant. The fold test of study A is positive while that of study B is inconlusive; this last result constrains the age of magnetization to syn- or post-Middle Devonian time (Acadian orogeny). What are the reasons that could explain the differences between these two studies? Clearly, they are of a lithological and structural nature. Van der Voo and Johnson (1985) did not list the values of their strikes and dips for the Dunn Point Formation; they mention only that the volcanic rocks are steeply dipping to the SE. In the present study both upright and overturned dips were sampled; this may partly explain the divergence between their positive fold test and the inconclusive fold test in the present study. Nevertheless, the mean direction (0°, −63°) of the in-situ *A* component of this study is not much different from that of van der Voo and Johnson (335°, −61°).

The occurrence of multi-component magnetization in this study is more at variance with van der Voo and Johnson's study. It was shown that component *B* is most probably older than component *A* and that component *B* is secondary. The difference in the number of components is certainly due to differing opaque mineralogy which, in turn, implies sampling either in different horizons of the Dunn Point Formation or different levels in the individual volcanic flows; in any event, this divergence appears to have its cause in a difference in lithology.

Paleopoles of component *A* (in-situ and tilt-corrected) are 163° E, 52° N ($dp = 7.1°$, $dm = 12.9°$) and 62° W, 0° ($dp = 13.5°$, $dm = 17.1°$), that of component *B* (in-situ) is 146° E, 4° N

($dp = 8.5°$, $dm = 12.2°$) and that of component *C* (in-situ) 10°W, 41°N ($dp = 16.2°$, $dm = 19°$). The pole position of component *B*, which is most probably older than component *A*, is located south of the Early Carboniferous poles obtained on the Deer Lake Group (Irving and Strong, 1984), the New Brunswick volcanics and red beds (Seguin et al., 1985) and the St. Lawrence granite (Irving and Strong, 1985). The age of this component is thus most probably Mid to Late Devonian. Mid to Late Devonian granites of the Ste.-Cecile–St. Sebastien region, which are located in the Gaspé–Connecticut Valley Anticlinorium of Quebec Appalachians (Seguin et al., 1981), yielded a comparable paleopole of 150°E, 18°N. However, the remanence direction of the Dunn Point Formation is reversed while that of the Ste.-Cécile–St.-Sébastien granites is normal. At that time, the paleolatitude of the Andean margin of North America was at approximately 18°S and that of Avalon at c. 33°S. Considering the angular errors involved, this means a gap of 1000–1500 km between the two landmasses. For the same period, the paleopole of cratonic North America was located at c. 120°E, 40°N (van der Voo, 1981). On the basis of this data set, it is plausible that the Avalon terrane(s) was (were) located south of the North American eastern margin at Mid–Late Devonian time.

The paleopole of component *B* is located close to those of Upper Jurassic–Lower Cretaceous age for North America (Harrison and Lindh, 1982). The pole of component *C* does not coincide with any known North American pole position.

The components of magnetization for the Dunn Point volcanics are not primary and cannot be used to solve the Silurian paleoreconstruction problem of Avalon. Indeed, the high paleolatitude obtained by van der Voo and Johnson (1985) is incompatible with the rich tropical to subtropical fauna of the Arisaig Group (Bambach, 1969). Where did Avalon terrane stand in Silurian time? Figure 8 shows the Silurian paleopoles obtained from the northern Appalachians (Dunn Point Formation excluded); clearly the Silurian paleopoles are scattered all over the diagram. Some systematization is thus needed. For this reason, the available Silurian (plus some Lower Devonian paleopoles) were subdivided into three regions: (1) cratonic, (2) Andean margin and (3) Avalon Zone. First, two poles are available for the cratonic region; they are located around 110°E, 25°N. Second, many poles are available on the Andean margin; they are clustered at 125°E and 20°N. Third, most of the Silurian paleopoles from the Avalon Zone (JE excepted) are situated in the southern latitudes between 5° and 35° and in about the some longitude range as the two other groups of poles.

8. Conclusion

Obviously, the Silurian rocks which constitute the Appalachian mountain chain come from very distant geographic localities. What can be obtained from the relative positions of the two sets of poles? The interpretation given to this observation is the following:

(1) existence of a sea (wide ocean) at Early Silurian time as indicated by the angular discordance between the two sets of poles (cratonic and Avalon Zone). This first geological situation explains the 35° latitude difference (minimum width of sea 3000 km) between cratonic and Avalon Zone poles. Slightly before this period, the Avalon plate broke up, the ignimbrites representing the initial rifting motion of the Avalon plate in Early Silurian time; and

(2) the second episode took place between Middle Silurian and Lower Devonian time. This implies closing of the sea and rapid APW. The APW is explained by the longitudinal difference between JC and CB-1 poles. By Early–Mid Devonian time, the cratonic and Andean poles are situated in the general area of 70–90°E and 0–30°N (Seguin and Gahé, 1985). By that time the sea is still not closed (1000–1500 km). In Upper Devonian time, the groups of poles should be characterized by a common APW only but this has not been demonstrated yet. During the Silurian period, the Andean margin poles are nearer to cratonic poles than Avalon poles which may mean that the closure of the ocean is at its beginning.

This scenario described above is one probable global tectonics model but many others are possi-

ble. A clear picture of paleoreconstitution of the northern Appalachians awaits further large-scale geological studies coupled to many additional paleomagnetic studies in Silurian and Devonian rocks of the Avalon terrane. At this stage, it is quite clear that insufficient paleomagnetic data are available to arrive at a definite solution for the provenance of Avalon terrane(s). One can hypothesize a common origin with Gondwana, Hercynian Europe or even Baltica on paleomagnetic ground, but such premature considerations are dangerous and may have adverse effects. In the last decade, too many hypotheses based on preliminary paleomagnetic arguments have proved to be a nuisance to the advancement of science in this specific field. We believe that a conservative approach for interpreting the paleomagnetic results is preferable if the credibility in this geophysical method is to be maintained by the geological community. With this approach in mind, it is concluded that the paleomagnetic results of the present paleomagnetic study cannot tell whether the Avalon Zone drifted separately and differently from cratonic North America in Silurian time. Taking into consideration the remagnetizations observed in the Dunn Point Formation, we cannot tell using the paleomagnetic results obtained from this study whether the Avalon Zone is a separate microcontinent or a part of northwestern Africa, Baltica or Hercynian Europe at Silurian time. This paleomagnetic study does not allow a comparison with other Silurian paleomagnetic data from the Avalon Zone and cannot tell whether this zone was constituted of one single terrane or a mosaic of smaller blocks. The present study tells us that by Middle to Late Devonian time the Avalon microcontinent was located south and close to the North American eastern margin.

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References

Bambach, R.K., 1969. Bivalvia of the Siluro–Devonian Arisaig Group, Nova Scotia. Ph.D. Thesis, Yale University, 376 pp. (unpublished).

Benson, D.G., 1974. Geology of the Antigonish Highlands, Nova Scotia. Geol. Surv. Can., Memoir 376, 92 pp.

Boucot, A.J., Dewey, J.F., Dineley, D.L., Fletcher, R., Fyson, W.K., Griffin, J.G., Hickox, C.F., McKerrow, W.S. and Zeigler, A.M., 1974. Geology of the Arisaig Area, Antigonish County, Nova Scotia, Geol. Soc. Am. Spec. Pap., 139, 1971 pp.

Briden, J.C. and Duff, B.A., 1981. Pre-Carboniferous Paleomagnetism of Europe North of the Alpine Orogenic Belt. In: M.W. McElhinny and D.A. Valencio (Editors), Paleoreconstruction of the Continents. Geodynamics Series, 2, Am. Geophys. Union; Geol. Soc. Am., pp. 137–149.

Brock, A. 1981. Paleomagnetism of Africa and Madagascar. In: M.W. McElhinny and D.A. Valencio (Editors), Paleoreconstruction of the Continents. Geodynamics Series, 2, Am. Geophys. Union; Geol. Soc. Am., pp. 65–76.

Deutsch, E.R. and Pätzold, R.R., 1976. Magnetism of basalt cores from the Nazca Plate and implications for magnetic anomaly interpretation. J. Geophys. Res., 81: 4188–4198.

Deutsch, E.R. and Pätzold, R.R., 1980. A Comment on: 'A detailed rock magnetic and opaque mineralogy study of the basalts from the Nazca Plate' by H.P. Johnson and J.M. Hall. Geophys. J., 6: 497–504.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. R. Soc. London, A217: 295–305.

French, A.N. and van der Voo, R., 1979. The Magnetization of the Rose Hill Formation at the classical site of Graham's fold test. J. Geophys. Res., 84: 7688–7696.

Fullager, P.D. and Bottino, M.L., 1968. Radiometric age of the volcanics at Arisaig, Nova Scotia, and the Ordovician–Silurian boundary. Can. J. Earth Sci., 5: 311–317.

Harrison, C.G.A. and Lindh, T., 1982. A Polar wandering curve for North America during the Mesozoic and Cenozoic. J. Geophys. Res., 87: 1903–1920.

Irving, E. and Strong, D.F., 1984. Paleomagnetism of the Early Carboniferous Deer Lake Group, western Newfoundland: no evidence for Carboniferous displacement of 'Acadia'. Earth Planet. Sci. Lett., 69: 379–391 (preprint).

Irving, E. and Strong, D.F., 1985. Paleomagnetism of rocks from Burin Peninsula, Newfoundland: hypothesis of late Paleozoic displacement of Acadia criticized. J. Geophys. Res., 90: 1949–1962.

Kent, D.V. and Opdyke, N.D., 1980. Paleomagnetism of Siluro–Devonian rocks from eastern Maine. Can. J. Earth Sci., 17: 1653–1665.

Keppie, J.D., Dostal, J. and Zentilli, M., 1978. Petrology of the Early Silurian Dunn Point and McGillivray Brook formations, Arisaig, Nova Scotia. Nova Scotia Department of Mines, Halifax, Paper 78–5, 20 pp.

Keppie, J.D., Dostal, J. and Zentilli, M., 1979. Early Silurian volcanic rocks at Arisaig, Nova Scotia. Can. J. Earth Sci., 16: 1635–1640.

Lapointe, P.L., 1979. Paleomagnetism and orogenic history of the Botwood Group and Mount Peyton batholith. Can. J. Earth Sci., 16: 866–876.

Lynnes, C.S. and van der Voo, R., 1984. Paleomagnetism of the Cambro–Ordovician McClure Mountain alkalic complex, Colorado. Earth Planet. Sci. Lett., 71: 163–172.

McElhinny, M.W., 1964. Statistical significance of the fold test in paleomagnetism. Geophys. J.R. Astron. Soc., 8: 338–340.

McKerrow, W.S. and Ziegler, A.M., 1972. Paleozoic oceans. Nature, 240: 92–94, 1972.

Murthy, G.S., Deutsch, E.R. and Pätzold, R.R., 1976. Inferences on the magnetic domain state of Leg 37 basalts. J. Geophys. Res., 81: 4199–4206.

O'Brien, S.J., Wardle, R.J. and King, A.F., 1983. The Avalon Zone: a Pan-African terrane in the Appalachian Orogen of Canada. Geol. J., 18: 195–222.

Perroud, H., van der Voo, R. and Bonhommet, N., 1984. Paleozoic evolution of the Armorica plate on the basis of paleomagnetic data. Geology, 12: 579–582.

Radhakrishnamurty, C. and Deutsch, E.R., 1974. Magnetic techniques for ascertaining the nature of iron oxide grains in basalts. J. Geophys., 40: 453–465.

Rao, K.V., Seguin, M.K.-, and Deutsch, E.R., 1981. Paleomagnetism of Siluro–Devonian and Cambrian granitic rocks from the Avalon Zone in Cape Breton Island, Nova Scotia. Can. J. Earth Sci., 18: 1187–1210.

Rast, N. and Skehan, J.W., 1983. The evolution of the Avalonian plate. Tectonophysics, 100: 257–286.

Roy, J.L. and Anderson, P., 1981. An investigation of the remanence characteristics of three sedimentary units of the Silurian Mascarene Group of New Brunswick. J. Geophys. Res., 86: 6351–6368.

Roy, J.L., Opdyke, N.D. and Irving, E., 1967. Further paleomagnetic results from the Bloomsburg Formation. J. Geophys. Res., 72: 5075–5086.

Roy, J.L., Anderson, P. and Lapointe, P.L., 1979. Paleomagnetic results from three rock units of New Brunswick and their bearing on the Lower Paleozoic tectonics of North America. Can. J. Earth. Sci., 16: 1210–1227.

Schenk, P.E., 1971. Southeastern Atlantic Canada, northwestern Africa, and continental drift. Can. J. Earth Sci., 8: 1218–1251.

Schenk, P.E., 1975. A regional synthesis of Canadian Appalachians. Marit. Sediments, 11: 17–24.

Seguin, M.K.-, 1980a. Paleomagnetism of Upper Silurian–Lower Devonian Units in the Gaspé–New Brunswick Region. Bulg. Geophys. J., 6: 86–101.

Seguin, M.K.-, 1980b. A case of multiphase magnetization in the Appalachians: the Weedon–Gould area. Ann. Geophys., 36: 73–84.

Seguin, M.K.- and Gahé, E., 1985. Paleomagnetism of Lower Devonian volcanics and Devonian dykes from northcentral New Brunswick, Canada. Phys. Earth Planet. Inter., 38: 262–276.

Seguin, M.K.- and Petryk, A.A., 1984. A Paleomagnetic Study of Upper Ordovician–Lower Silurian Strata of Anticosti Island. Northeast. Geol., 6: 25–37.

Seguin, M.K.- and Petryk, A.A., 1985. Upper Ordovician–Lower Silurian sedimentary sequences of Anticosti Island; a paleomagnetic investigation, Program with Abstracts. Geol. Assoc. Can., A56.

Seguin, M.K.-, Rao, K.V. and Arnal, P., 1981a. Paleomagnetic Study of Cambrian Postdam Group Sandstones, St. Lawrence Lowlands, Quebec. Earth Planet. Sci. Lett., 55: 433–449.

Seguin, M.K.-, Rao, K.V., Deutsch, E.R. and Gahé, E., 1981b. Paleomagnetism of Silurian Dunn Point volcanics, Avalon Zone, Nova Scotia. European Geophys. Soc., 8th annual meeting, Uppsala, Sweden, 102 (abstract).

Seguin, M.K.-, Rao, K.V. and Pineault, R., 1982. Paleomagnetism of Devonian Rocks From Ste.Cécile–St.-Sébastien Region, Quebec Appalachians. J. Geophys. Res., 87: 7853–7864.

Seguin, M.K.-, Singh, A. and Fyffe, L., 1985. New Paleomagnetic data from Carboniferous volcanics and Red Beds from Central New Brunswick. Geophys. Res. Lett., 12: 81–84.

Skehan, J.W. and Rast, N., 1983. Relationship between Precambrian and lower Paleozoic rocks of southeastern New England and other North Atlantic Avalonian terrains. In: P.E. Schenk (Editor), Regional Trends in the Geology of the Appalachian–Caledonian–Hercynian–Mauritanide Orogen, NATO ASI Series C, 16: 131–162.

Van der Voo, R., 1981. Paleomagnetism of North America: a brief review. In: M.W. McElhinny and P.A. Valencio (Editors), Paleoreconstruction of the Continents. Geodynamics Series, 2, Am. Geophys. Union; Geol. Soc. Am., pp. 159–176.

Van der Voo, R. and Johnson, R.J.E., 1985. Paleomagnetism of the Dunn Point Formation (Nova Scotia): high Paleolatitudes for the Avalon terrane in the Late Ordovician. Geophys. Res. Lett., 12: 337–340.

Williams, H. and Hatcher Jr., R.D., 1983. Appalachian suspect terranes, In: R.D. Hatcher, H. Williams and I. Zietz (Editors), Contribution to the Tectonics and Geophysics of Mountain Chains. Geol. Soc. Am. Mem., 158: 33–54.

Wilson, J.T., 1966. Did the Atlantic close and then reopen? Nature, 211: 676–681.